Course Notes

for (CSED490V) Foundations of Continuous Computation

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Notation and Conventions

General notation and writing follow *(MATH000) Mathematical Language*.

Notation	Meaning
$\mathbb{R}_C$	Computable real numbers
$\mathcal{A}$	Oracle algorithm
$\gamma_{E,M}$	Floating-point representation map
$\mathbb{F}_{E,M}$	Finite floating-point values
$\overline{\mathbb{F}}_{E,M}$	Extended floating-point values
$e_{\min}$	Minimum normal exponent
$e_{\max}$	Maximum normal exponent
$F_{\max}$	Largest finite floating-point value
$K_{E,M}$	Overflow threshold
$\mathrm{rnd}_{E,M}$	Round to nearest, ties to even
$\oplus$	Rounded addition
$\ominus$	Rounded subtraction
$\otimes$, $\odot$	Rounded multiplication
$\oslash$, $\oslash$	Rounded division
fma	Fused multiply-add
$\mathrm{err}_{\mathrm{abs}}$	Absolute error
$\mathrm{err}_{\mathrm{rel}}$	Relative error
ulp	Unit in the last place
$\mathrm{err}_{\mathrm{ulp}}$	ULP error
e_a	Exponent of a nonzero floating-point value a
fast2sum	Error-free summation algorithm
σ	Input valuation
Σ	Set of admissible input valuations
$[\![e]\!]$	Floating-point semantics of e
$[\![e]\!]_{\mathrm{r}}$	Real-valued semantics of e
$\mathrm{range}(e)$	Range enclosure over Σ
$\mathrm{err}(e)$	Absolute-error bound over Σ
ε	Machine epsilon (unit roundoff)

1 Computable Reals

Definition 1.1. Borel Computability
A real number r is *Borel-computable* if there exists a computable sequence $(q_n)_{n\in\mathbb{N}_0} \subseteq \mathbb{Q}$ such that

$$|r - q_n| < 2^{-n} \qquad (n \in \mathbb{N}_0).$$

Definition 1.2. Turing Computability
A nonnegative real number r is *Turing-computable* if it has a binary expansion

$$r = a + \sum_{i=0}^{\infty} b_i 2^{-(i+1)}, \qquad a \in \mathbb{N}_0, \quad b_i \in \{0,1\},$$

for which $i \mapsto b_i$ is computable. A negative real number is Turing-computable when its absolute value is Turing-computable. For dyadic rationals we may use the terminating binary expansion.

Theorem 1.3. Equivalence of Computability Definitions
For $r \in \mathbb{R}$, the following are equivalent:

(i) r is Borel-computable.

(ii) r is Turing-computable.

Proof. We may consider $r \geq 0$: changing the sign of a computable rational approximation is effective, and the negative case of binary computability is defined using the absolute value.

Suppose first that r is Turing-computable, with integer part a and computable bits b_i. Set

$$q_n = a + \sum_{i=0}^{n} b_i 2^{-(i+1)} \in \mathbb{Q}.$$

The sum is computable and

$$0 \leq r - q_n \leq \sum_{i=n+1}^{\infty} 2^{-(i+1)} = 2^{-(n+1)} < 2^{-n}.$$

Thus r is Borel-computable.

Conversely, suppose $(q_m)_{m\in\mathbb{N}_0}$ computes r with $|q_m - r| < 2^{-m}$. Fix the integer part $a = \lfloor r \rfloor$ as part of the existential description of a binary expansion. We distinguish two cases.

If r is dyadic, it admits an eventually zero binary expansion. Its finite sequence of nonzero digits can be built into an algorithm, and hence r is Turing-computable.

Now assume r is not dyadic. Fix $n \in \mathbb{N}_0$. The non-dyadic number r lies strictly between two consecutive grid points of mesh $2^{-(n+1)}$. Search over $m \in \mathbb{N}_0$ until the rational interval

$$I_m = (q_m - 2^{-m},\, q_m + 2^{-m})$$

is contained strictly in a single grid cell

$$\left(\frac{k}{2^{n+1}}, \frac{k+1}{2^{n+1}}\right), \qquad k \in \mathbb{N}_0.$$

Containment is decidable by rational arithmetic: for a given m there is at most one candidate k, which can be found by integer division. The search terminates, since r has positive distance from both endpoints of its grid cell and $q_m \to r$ with an effective error bound. Moreover $r \in I_m$, so the returned k satisfies $k = \lfloor 2^{n+1} r \rfloor$. The nth binary digit of r is therefore

$$b_n = k \bmod 2.$$

This procedure computes each bit, completing the proof. □

Remark. The two definitions characterize the same set of real numbers, but their representations do not admit symmetric *uniform* conversions. Given a computable binary expansion one can effectively obtain rational approximations. In general there is no algorithm that takes an arbitrary rational-approximation procedure and uniformly produces its binary digits: the latter operation must resolve boundary cases at dyadic rationals. This does not contradict the existence of a computable binary expansion for each Borel-computable real.

Definition 1.4. Computable Real Numbers
The set of *computable real numbers* is

$$\mathbb{R}_C := \{r \in \mathbb{R} \mid r \text{ is computable}\}.$$

By the preceding theorem, either notion of computability can be used.

Proposition 1.5. Closure under Arithmetic
The set $\mathbb{R}_C$ is a subfield of $\mathbb{R}$. In particular,

$$x, y \in \mathbb{R}_C \implies x + y,\ x - y,\ xy \in \mathbb{R}_C, \qquad y \neq 0 \implies \frac{x}{y} \in \mathbb{R}_C.$$

Proposition 1.6. Real Algebraic Closure
If $p \in \mathbb{R}_C[X]$ is nonzero, then every real root of p lies in $\mathbb{R}_C$. In particular, $\mathbb{R}_C$ is a real-closed field.

Proposition 1.7. Computability of π and e
The real numbers π and e are computable.

Proof. The alternating arctangent series gives

$$\frac{\pi}{4} = \arctan 1 = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1}.$$

For $m \in \mathbb{N}_0$ put

$$p_m = 4 \sum_{k=0}^{m} \frac{(-1)^k}{2k+1} \in \mathbb{Q}.$$

The alternating-series estimate yields

$$|\pi - p_m| \leq \frac{4}{2m+3}.$$

Choosing $m = 2^{n+1}$ gives $|\pi - p_m| < 2^{-n}$ for every $n \in \mathbb{N}_0$. The finite sum p_m is computable from n.

The computability of $e = \exp(1)$ follows from the computability of $\exp\colon \mathbb{R} \to \mathbb{R}$ proved in Theorem 2.7. □

Proposition 1.8. Countability of the Computable Reals
The set $\mathbb{R}_C$ is countably infinite. Consequently, there exist noncomputable real numbers.

Proof. There are countably many finite descriptions of algorithms, and every computable real has at least one such description. Hence $\mathbb{R}_C$ is at most countable. On the other hand, every rational number is computable via its constant approximation sequence, so $\mathbb{Q} \subseteq \mathbb{R}_C$ and $\mathbb{R}_C$ is infinite. Since $\mathbb{R}$ is uncountable, $\mathbb{R} \setminus \mathbb{R}_C \neq \varnothing$. □

Fact 1.9. Hilbert's Tenth Problem
There is no algorithm that decides, for every multivariate polynomial with integer coefficients, whether it has an integer zero. In particular, the set of integer polynomials admitting an integer solution is not decidable.

Proposition 1.10. A Noncomputable Real
Let P be the set of all integer-coefficient polynomials in finitely many variables, and let $\phi\colon P \to \mathbb{N}$ be a computable injection. Write P_{root} for the polynomials having an integer zero. Then

$$r \coloneqq \sum_{p \in P} 2^{-\phi(p)}\, \mathbb{1}_{P_{\text{root}}}(p)$$

is a noncomputable real in $[0, 1]$.

Proof. Injectivity gives the binary coefficient at $2^{-\phi(p)}$:

$$b_{\phi(p)} = \mathbb{1}_{P_{\text{root}}}(p).$$

Both P_{root} and $P \setminus P_{\text{root}}$ are infinite, so the expansion has infinitely many zeros and ones; in particular, r is not dyadic. If $r \in \mathbb{R}_C$, its unique binary expansion is computable by the equivalence theorem. Hence

$$p \longmapsto b_{\phi(p)} = \mathbb{1}_{P_{\text{root}}}(p)$$

would decide Hilbert's Tenth Problem, a contradiction. □

2 Computable Functions

Definition 2.1. Computable Function
Let $D \subseteq \mathbb{R}$. A function $f\colon D \to \mathbb{R}$ is *computable* if there exists an oracle algorithm $\mathcal{A}$ with the following property. For every $x \in D$, every function $\alpha\colon \mathbb{N}_0 \to \mathbb{Q}$ satisfying

$$|\alpha(i) - x| < 2^{-i} \qquad (i \in \mathbb{N}_0),$$

and every precision $n \in \mathbb{N}_0$, the algorithm $\mathcal{A}$, given access to α and the input n, halts and returns a rational number $\mathcal{A}(\alpha, n)$ such that

$$|\mathcal{A}(\alpha, n) - f(x)| < 2^{-n}.$$

The algorithm may query finitely many values $\alpha(i)$ during a computation. The definition extends componentwise to functions between subsets of $\mathbb{R}^k$ and $\mathbb{R}^m$.

Remark. A name α need not be computable: $\mathcal{A}$ must work for *every* rational name, even one naming a noncomputable real. Unlike a binary-digit oracle, rational approximation does not require a uniform decision at dyadic boundaries (cf. Section 1).

In particular, every computable $f\colon D \to \mathbb{R}$ satisfies $f(D \cap \mathbb{R}_C) \subseteq \mathbb{R}_C$.

Example 2.2. Basic Computable Functions
For $r \in \mathbb{R}_C$, both $x \mapsto x + r$ and $x \mapsto rx$ are computable. Other examples are $|x|$, e^x, $\sin x$, $\sqrt{x}$ (for $x \geq 0$), and $\log x$ (for $x > 0$).

In several variables, $+$, $-$, multiplication, min, and max are computable; division is computable when the divisor is nonzero.

Theorem 2.3. Closure under Composition
Let $f\colon D \to E$ and $g\colon E \to \mathbb{R}^m$ be computable, where $D \subseteq \mathbb{R}^k$ and $E \subseteq \mathbb{R}^\ell$. Then $g \circ f$ is computable.

Corollary 2.4. Computable Function Operations
If $f, g\colon D \to \mathbb{R}$ are computable, then so are

$$|f|, \quad f + g, \quad fg, \quad \min(f, g), \quad \max(f, g), \quad \exp \circ f.$$

Moreover, f/g is computable on $\{x \in D \mid g(x) \neq 0\}$.

Theorem 2.5. Computability Implies Continuity
Every computable function $f\colon D \to \mathbb{R}^m$, with $D \subseteq \mathbb{R}^k$, is continuous on D (with the relative topology).

Proof. Fix $\mathbf{x} \in D$ and $\varepsilon > 0$; choose $n \in \mathbb{N}_0$ with $2^{1-n} < \varepsilon$. Choose a rational name $\alpha\colon \mathbb{N}_0 \to \mathbb{Q}^k$ of $\mathbf{x}$ with slack:

$$\|\alpha(i) - \mathbf{x}\|_\infty < 2^{-(i+2)} \qquad (i \in \mathbb{N}_0).$$

At precision n, the algorithm queries a finite set of indices I. For some $\delta > 0$, every $\mathbf{y} \in D$ with $\|\mathbf{x} - \mathbf{y}\|_\infty < \delta$ also admits a name agreeing with α on I. The algorithm therefore returns the same $\mathbf{q} \in \mathbb{Q}^m$ on both names, whence

$$\|f(\mathbf{x}) - f(\mathbf{y})\|_\infty \leq \|f(\mathbf{x}) - \mathbf{q}\|_\infty + \|\mathbf{q} - f(\mathbf{y})\|_\infty < 2^{1-n} < \varepsilon.$$

Thus f is continuous at $\mathbf{x}$. □

Remark. Continuity is necessary, not sufficient, for computability: if $c \in \mathbb{R} \setminus \mathbb{R}_C$, the constant function $x \mapsto c$ is continuous but not computable. In particular, a function with a discontinuity at a point of its domain cannot be computable there.

Example 2.6. Discontinuous Functions and Their Domains
The step function $x \mapsto \mathbb{1}_{\{x>0\}}$ is not computable on $\mathbb{R}$ but is computable on $\mathbb{R} \setminus \{0\}$: refine an input approximation until its sign is certified. The same principle applies to strict comparison on $x \neq y$.

A set-valued extension, outside the single-valued definition above, can instead allow both outputs at the discontinuity:

$$\operatorname{sgn}'(x) = \begin{cases} \{1\}, & x > 0, \\ \{0\}, & x < 0, \\ \{0,1\}, & x = 0. \end{cases}$$

Theorem 2.7. Computability of the Exponential Function
The function $\exp\colon \mathbb{R} \to \mathbb{R}$ *is computable.*

Proof. First let $x \in [-\frac{1}{2}, \frac{1}{2}]$, let α be a rational name of x, and fix $n \in \mathbb{N}_0$. Set $m = d = n + 3$ and

$$\widehat{x} = \alpha(m), \qquad \widehat{y} = P_d(\widehat{x}), \qquad P_d(t) = \sum_{j=0}^{d} \frac{t^j}{j!}.$$

The rational $\widehat{y}$ is computed by finite arithmetic. Since $|x - \widehat{x}| < 2^{-m}$ and $|\widehat{x}| < 1$, the mean value theorem and $e < 3$ give

$$|e^x - e^{\widehat{x}}| < 3 \cdot 2^{-m}.$$

For $d \geq 3$, the Taylor remainder satisfies

$$|e^{\widehat{x}} - P_d(\widehat{x})| \leq \sum_{j=d+1}^{\infty} \frac{1}{j!} \leq \sum_{j=d+1}^{\infty} 2^{-j} = 2^{-d}.$$

Consequently,

$$|e^x - \widehat{y}| < (3+1)2^{-(n+3)} < 2^{-n}.$$

For arbitrary $x \in \mathbb{R}$, one can effectively choose $k \in \mathbb{N}_0$ with $|x|/2^k \leq \frac{1}{2}$. Write $u = e^{x/2^k}$ and $N = 2^k$. The preceding algorithm computes $u \in (0, 2)$ to arbitrary precision. For any rational v with $|u - v| < 1$,

$$|u^N - v^N| \leq N 3^{N-1} |u - v|.$$

Thus a sufficiently accurate rational approximation v gives the required approximation v^N to $e^x = u^N$. □

Theorem 2.8. Noncomputability of Strict Comparison
The strict-comparison map $\mathbb{R}^2 \to \{0,1\}$ *defined by*

$$(x, y) \mapsto \begin{cases} 1, & x < y, \\ 0, & x \geq y \end{cases}$$

is not computable.

Proof. Suppose an algorithm computes strict comparison. Because its values lie in $\{0,1\}$, requesting an approximation with error below $1/4$ and testing whether the result exceeds $1/2$ yields an algorithm returning the exact bit. Give this decision algorithm the two names

$$\alpha(i) = 0, \qquad \beta(i) = 2^{-(i+1)} \qquad (i \in \mathbb{N}_0),$$

both of which represent the real number 0. Hence the algorithm must halt and output 0. During this run, only finitely many indices of β are queried; let $N \in \mathbb{N}_0$ exceed or equal all of them. Define another rational name by

$$\beta'(i) = \begin{cases} 2^{-(i+1)}, & i \le N, \\ 2^{-(N+1)}, & i > N. \end{cases}$$

For $i \le N$, its distance from $2^{-(N+1)}$ is strictly smaller than 2^{-i}, and for $i > N$ the distance is zero. Thus β' is a valid name of $y' = 2^{-(N+1)} > 0$. The algorithm cannot distinguish (α, β') from (α, β) during its finite computation; it still returns 0. But strict comparison at $(0, y')$ equals 1, a contradiction. □

Theorem 2.9. Comparison of Unequal Real Numbers
Strict comparison is computable on

$$\{(x, y) \in \mathbb{R}^2 \mid x \neq y\}.$$

Remark. If $x \neq y$, sufficiently accurate rational approximations eventually certify their order. Without this promise, neither equality nor strict comparison is decidable on all of $\mathbb{R}^2$: their indicator maps are discontinuous along $x = y$.

3 Floating-Point Arithmetic

Definition 3.1. Floating-Point Representation
Fix $E \ge 2$ and $M \ge 1$. A floating-point bit string consists of a sign bit s, E exponent bits, and M fraction bits. Let

$$k = (e_1 \cdots e_E)_2, \qquad t = (0.m_1 \cdots m_M)_2 = \sum_{j=1}^{M} m_j 2^{-j}, \qquad b_E = 2^{E-1} - 1.$$

The interpretation map $\gamma_{E,M} \colon \{0,1\}^{1+E+M} \to \mathbb{R} \cup \{-\infty, +\infty, \mathrm{NaN}\}$ is defined by

$$\gamma_{E,M}(s, e_1, \ldots, e_E, m_1, \ldots, m_M) = \begin{cases} (-1)^s 2^{k-b_E}(1+t), & 1 \le k \le 2^E - 2, \\ (-1)^s 2^{1-b_E} t, & k = 0, \\ (-1)^s \infty, & k = 2^E - 1,\ t = 0, \\ \mathrm{NaN}, & k = 2^E - 1,\ t \neq 0. \end{cases}$$

The first case represents *normal* values. The second represents *subnormal* values when $t \neq 0$ and zero when $t = 0$; the remaining cases represent infinities and *not a number* (NaN). The normal exponent bounds are

$$e_{\min} = 1 - b_E = 2 - 2^{E-1}, \qquad e_{\max} = 2^E - 2 - b_E = b_E.$$

Write

$$\overline{\mathbb{F}}_{E,M} = \operatorname{im} \gamma_{E,M}, \qquad \mathbb{F}_{E,M} = \overline{\mathbb{F}}_{E,M} \setminus \{-\infty, +\infty, \mathrm{NaN}\}.$$

Thus $\mathbb{F}_{E,M} \subseteq \mathbb{R}$ is the set of finite floating-point values.

Remark. The two signed-zero bit strings both represent $0 \in \mathbb{F}_{E,M}$, while several distinct bit strings represent NaN. The map $\gamma_{E,M}$ identifies these representations as values. In the usual binary32 and binary64 formats, $(E, M) = (8, 23)$ and $(11, 52)$, respectively.

Definition 3.2. Rounding
Let

$$F_{\max} = (2 - 2^{-M})2^{b_E}, \qquad K_{E,M} = (2 - 2^{-M-1})2^{b_E}.$$

Here $F_{\max}$ is the largest finite floating-point value and $K_{E,M}$ is the positive overflow threshold. The *round-to-nearest, ties-to-even* map $\mathrm{rnd}_{E,M} \colon \mathbb{R} \to \overline{\mathbb{F}}_{E,M}$ is defined by

$$\mathrm{rnd}_{E,M}(x) = \begin{cases} -\infty, & x \le -K_{E,M}, \\ \operatorname*{arg\,min}_{y \in \mathbb{F}_{E,M}} |x - y|, & |x| < K_{E,M}, \\ +\infty, & x \ge K_{E,M}. \end{cases}$$

If two finite values minimize the distance, choose the one whose least significant stored significand bit is even. In particular, halfway cases near 0 round to 0; the chosen value always exists and is unique.

Proposition 3.3. Monotonicity of Rounding
For $x, y \in \mathbb{R}$,

$$x \le y \quad \Longrightarrow \quad \mathrm{rnd}_{E,M}(x) \le \mathrm{rnd}_{E,M}(y),$$

with the usual order on $\mathbb{R} \cup \{-\infty, +\infty\}$.

Definition 3.4. Floating-Point Operations
For $a, b, c \in \mathbb{F}_{E,M}$, define rounded arithmetic by

$$\begin{aligned} a \oplus b &= \mathrm{rnd}_{E,M}(a + b), \quad a \ominus b = \mathrm{rnd}_{E,M}(a - b), \\ a \otimes b &= \mathrm{rnd}_{E,M}(ab), \qquad a \oslash b = \mathrm{rnd}_{E,M}(a/b) \quad (b \neq 0). \end{aligned}$$

The secondary notations are $\odot$ for multiplication and $\oslash$ for division. *Fused multiply-add* performs only one rounding:

$$\mathrm{fma}(a, b, c) = \mathrm{rnd}_{E,M}(ab + c).$$

These operations have values in $\overline{\mathbb{F}}_{E,M}$; operations involving infinities, NaN, or signed-zero bit patterns are outside the present finite-input definitions.

Remark. Algebraic Properties. Rounded addition and multiplication are commutative, with identities 0 and 1, but generally neither associative nor distributive. For $(E, M) = (3, 2)$, ties-to-even gives

$$\begin{aligned} (1 \oplus \tfrac{1}{8}) \oplus (-1) = 0 \neq \tfrac{1}{8} = 1 \oplus (\tfrac{1}{8} \oplus (-1)), \\ 2 \otimes (\tfrac{3}{4} \otimes \tfrac{1}{8}) = \tfrac{1}{4} \neq \tfrac{3}{16} = (2 \otimes \tfrac{3}{4}) \otimes \tfrac{1}{8}, \\ 3 \otimes (1 \oplus \tfrac{3}{2}) = 8 \neq 7 = (3 \otimes 1) \oplus (3 \otimes \tfrac{3}{2}). \end{aligned}$$

Every $a \in \mathbb{F}_{E,M}$ has the unique additive inverse $-a$, since $a \oplus b = 0 \iff b = -a$ (all finite values are integer multiples of the smallest subnormal spacing). Multiplicative inverses need not exist or be unique. In the same format,

$$\tfrac{3}{2} \otimes \tfrac{5}{8} = 1 = \tfrac{3}{2} \otimes \tfrac{3}{4}, \qquad \tfrac{1}{16} \otimes b \neq 1 \quad (b \in \mathbb{F}_{3,2}).$$

Moreover, rounded operations may overflow to $\pm\infty$, outside $\mathbb{F}_{E,M}$; hence they do not form a field.

4 Floating-Point Errors

Fix the format (E, M) of Section 3. In this section write $\mathbb{F} = \mathbb{F}_{E,M}$ and $\mathrm{rnd} = \mathrm{rnd}_{E,M}$, and let $F_{\max}$, $e_{\min}$, and $e_{\max}$ retain their earlier meanings. For $x \in \mathbb{R}$, call x in the *normal range* when $2^{e_{\min}} \le |x| \le F_{\max}$ and in the *subnormal range* when $|x| < 2^{e_{\min}}$.

Definition 4.1. Absolute and Relative Errors
For an exact value $x \in \mathbb{R}$ and an approximation $\widehat{x} \in \mathbb{F}$, set

$$\mathrm{err}_{\mathrm{abs}}(x, \widehat{x}) \coloneqq |x - \widehat{x}|,$$
$$\mathrm{err}_{\mathrm{rel}}(x, \widehat{x}) \coloneqq \frac{|x - \widehat{x}|}{|x|} \qquad (x \neq 0).$$

Definition 4.2. Unit in the Last Place
The *unit in the last place* is the function $\mathrm{ulp}\colon \mathbb{R} \to \mathbb{R}_{>0}$ defined by

$$\mathrm{ulp}(x) \coloneqq \begin{cases} 2^{e-M}, & 2^e \le |x| < 2^{e+1} \text{ for some integer } e \ge e_{\min}, \\ 2^{e_{\min}-M}, & |x| < 2^{e_{\min}}. \end{cases}$$

The *ULP error* of $\widehat{x} \in \mathbb{F}$ is

$$\mathrm{err}_{\mathrm{ulp}}(x, \widehat{x}) \coloneqq \frac{|x - \widehat{x}|}{\mathrm{ulp}(x)}.$$

Thus ULP error measures the discrepancy in units of the local floating-point spacing (within a common binade). The *machine epsilon* (unit roundoff) of the format is

$$\varepsilon \coloneqq \tfrac{1}{2}\,\mathrm{ulp}(1) = 2^{-M-1}.$$

Proposition 4.3. Relative and ULP Errors
Let $x \in \mathbb{R} \setminus \{0\}$ and $\widehat{x} \in \mathbb{F}$. Then

$$\mathrm{err}_{\mathrm{ulp}}(x, \widehat{x}) \le \frac{\mathrm{err}_{\mathrm{rel}}(x, \widehat{x})}{\varepsilon}.$$

If x is in the normal range, then also

$$\mathrm{err}_{\mathrm{ulp}}(x, \widehat{x}) \ge \frac{\mathrm{err}_{\mathrm{rel}}(x, \widehat{x})}{2\varepsilon}.$$

Proof. The definitions give, without dividing by an error that might vanish,

$$\mathrm{err}_{\mathrm{ulp}}(x, \widehat{x}) = \mathrm{err}_{\mathrm{rel}}(x, \widehat{x}) \frac{|x|}{\mathrm{ulp}(x)}.$$

For $2^e \le |x| < 2^{e+1}$ with $e \ge e_{\min}$,

$$2^M \le \frac{|x|}{\mathrm{ulp}(x)} < 2^{M+1} = \frac{1}{\varepsilon}.$$

In the subnormal range, the same ratio is at most 2^M; this proves the first inequality. In the normal range its lower bound is $2^M = 1/(2\varepsilon)$, proving the second.

The normal-range hypothesis cannot be dropped. Indeed, for $x = 2^{e_{\min}-M}$ and $\widehat{x} = 0$, both the relative and ULP errors equal 1, but $1/(2\varepsilon) = 2^M > 1$. □

Proposition 4.4. Rounding Error Bounds
For round-to-nearest, ties-to-even, the following estimates hold:

(i) If x is in the normal range, then

$$\mathrm{err}_{\mathrm{rel}}(x, \mathrm{rnd}(x)) \le \varepsilon.$$

(ii) If x is in the subnormal range, then

$$\mathrm{err}_{\mathrm{abs}}(x, \mathrm{rnd}(x)) \le 2^{e_{\min}}\varepsilon.$$

(iii) If $|x| \le F_{\max}$, then

$$\mathrm{err}_{\mathrm{ulp}}(x, \mathrm{rnd}(x)) \le \tfrac{1}{2}.$$

Proof. For $2^e \le |x| < 2^{e+1}$ in the normal range, consecutive floating-point values are separated by at most 2^{e-M} locally. Thus nearest rounding gives

$$\frac{|x - \mathrm{rnd}(x)|}{|x|} \le \frac{2^{e-M-1}}{2^e} = \varepsilon.$$

In the subnormal range the spacing is $2^{e_{\min}-M}$, so

$$|x - \mathrm{rnd}(x)| \le \tfrac{1}{2}2^{e_{\min}-M} = 2^{e_{\min}}\varepsilon.$$

Finally, for $|x| \le F_{\max}$, nearest rounding is within half the local spacing $\mathrm{ulp}(x)$, giving (iii). □

Definition 4.5. Underflow and Overflow
For an exact nonzero arithmetic result z, the *underflow range* is $0 < |z| < 2^{e_{\min}}$. A result beyond the finite range satisfies $|z| > F_{\max}$; under round-to-nearest, actual *overflow to infinity* occurs when $|z| \ge K_{E,M}$. These two upper thresholds need not coincide.

Theorem 4.6. Error of Floating-Point Operations
Let $a, b \in \mathbb{F}$ and choose corresponding operations

$$* \in \{+, -, \times, \div\}, \qquad \circledast \in \{\oplus, \ominus, \otimes, \oslash\},$$

with $b \neq 0$ for division. If $z = a * b$ satisfies $|z| \le F_{\max}$, then

$$a \circledast b = z(1+\delta) + \delta', \qquad |\delta| \le \varepsilon, \quad |\delta'| \le 2^{e_{\min}}\varepsilon,$$

where one may choose $\delta' = 0$ in the normal range and $\delta = 0$ in the subnormal range (including $z = 0$). For addition and subtraction, one may always choose $\delta' = 0$.

Proof. By definition, $a \circledast b = \mathrm{rnd}(z)$. In the normal range, the relative rounding bound yields $\mathrm{rnd}(z) = z(1+\delta)$ with $|\delta| \le \varepsilon$. In the subnormal range, the absolute bound yields $\mathrm{rnd}(z) = z + \delta'$ with $|\delta'| \le 2^{e_{\min}}\varepsilon$.

For $* \in \{+, -\}$, every element of $\mathbb{F}$ is an integer multiple of $2^{e_{\min}-M}$, so $a \pm b$ is also such a multiple. If this exact sum or difference is in the subnormal range, it is representable and rounding is exact. Thus $\delta' = 0$ in both ranges. □

Example 4.7. Error in Repeated Multiplication
Let $a, b, c \in \mathbb{F} \setminus \{0\}$, and suppose both intermediate exact products are in the normal range and do not overflow. By the preceding theorem, for some $|\delta_1|, |\delta_2| \le \varepsilon$,

$$\begin{aligned}(a \otimes b) \otimes c &= \big(ab(1+\delta_1)c\big)(1+\delta_2)\\ &= abc(1+\delta_1)(1+\delta_2).\end{aligned}$$

Consequently, with $x = abc$ and $\widehat{x} = (a \otimes b) \otimes c$,

$$\begin{aligned}\mathrm{err}_{\mathrm{rel}}(x, \widehat{x}) &\le |\delta_1| + |\delta_2| + |\delta_1\delta_2| \le 2\varepsilon + \varepsilon^2,\\ \mathrm{err}_{\mathrm{ulp}}(x, \widehat{x}) &\le 2 + \varepsilon.\end{aligned}$$

To first order the relative error is at most 2ε.

Example 4.8. Error in Repeated Addition
Let $a, b, c \in \mathbb{F}$ and set $s = a+b$, $u = a+b+c \neq 0$. Assume $|s|, |(a \oplus b)+c| \leq F_{\max}$. For rounded addition, the theorem gives $\delta' = 0$ even for subnormal results; hence

$$\begin{aligned}(a \oplus b) \oplus c &= \big((a+b)(1+\delta_1)+c\big)(1+\delta_2) \\ &= u(1+\delta_2) + s\delta_1(1+\delta_2), \qquad |\delta_1|, |\delta_2| \leq \varepsilon.\end{aligned}$$

Writing $x = u$ and $\widehat{x} = (a \oplus b) \oplus c$, we obtain

$$\begin{aligned}\mathrm{err}_{\mathrm{rel}}(x, \widehat{x}) &\leq \varepsilon \left(1 + \frac{|a+b|}{|a+b+c|}\right) + \varepsilon^2 \frac{|a+b|}{|a+b+c|}, \\ \mathrm{err}_{\mathrm{ulp}}(x, \widehat{x}) &\leq 1 + \frac{|a+b|}{|a+b+c|} + \varepsilon \frac{|a+b|}{|a+b+c|}.\end{aligned}$$

The ratio $|a+b|/|a+b+c|$ can amplify the first rounding error.

For example, when $2^{M+1} \in \mathbb{F}$, take $a = 1$, $b = 2^{M+1}$, and $c = -2^{M+1}$. Ties-to-even yields

$$(a \oplus b) \oplus c = 0, \qquad a+b+c = 1,$$

with relative error 1 and ULP error 2^M.

Remark. Catastrophic Cancellation. When large, nearly equal quantities cancel, an error previously negligible relative to the intermediate values can dominate the final result. The preceding example loses the small term a in the first rounded addition.

The same phenomenon can reveal *pre-existing input errors*. If $\widehat{x}, \widehat{y} \in \mathbb{F}$ approximate $x, y \in \mathbb{R}$, then even when their subtraction is exact,

$$|(\widehat{x} - \widehat{y}) - (x - y)| \leq |\widehat{x} - x| + |\widehat{y} - y|.$$

For $x \neq y$, dividing by $|x - y|$ gives the relative error; dividing by $\mathrm{ulp}(x - y)$ gives the ULP error. Both can be large when $x \approx y$, even if the subtraction itself is exact.

Example 4.9. Stable Evaluation of $a^2 - b^2$
For $a, b \in \mathbb{F}$, $a^2 \neq b^2$, compare

$$u = (a \otimes a) \ominus (b \otimes b), \qquad v = (a \ominus b) \otimes (a \oplus b).$$

Assume that the operations do not overflow or underflow and that the usual relative-error model applies to all intermediate results. Writing $|\delta_i| \leq \varepsilon$ and retaining terms through first order gives

$$\begin{aligned}u &= (a^2(1+\delta_1) - b^2(1+\delta_2))(1+\delta_3), \\ \frac{|u - (a^2 - b^2)|}{|a^2 - b^2|} &\leq \left(1 + \frac{a^2 + b^2}{|a^2 - b^2|}\right)\varepsilon + O(\varepsilon^2), \\ v &= (a-b)(a+b)(1+\delta_1)(1+\delta_2)(1+\delta_3), \\ \frac{|v - (a^2 - b^2)|}{|a^2 - b^2|} &\leq 3\varepsilon + O(\varepsilon^2).\end{aligned}$$

For a normal exact result, the corresponding ULP-error bounds are $1 + (a^2 + b^2)/|a^2 - b^2| + O(\varepsilon)$ and $3 + O(\varepsilon)$, respectively. The first can be arbitrarily large when $|a| \approx |b|$; the factored expression avoids subtracting two rounded squares.

Remark. For $r, r' \in \mathbb{R}$ with $\mathrm{err}_{\mathrm{ulp}}(r, r') \le 1$ and finite $\mathrm{rnd}(r')$,

$$\mathrm{err}_{\mathrm{ulp}}(r, \mathrm{rnd}(r')) \le \mathrm{err}_{\mathrm{ulp}}(r, r') + \tfrac{1}{2}.$$

Indeed, if $\mathrm{ulp}(r') \le \mathrm{ulp}(r)$, the triangle inequality and nearest rounding suffice. Otherwise r' lies across a power-of-two boundary from r. The one-ULP hypothesis forces r' to round back to that boundary, closer to r than r' was. The estimate thus holds in both cases.

Example 4.10. Stable Quadratic Roots
For $a \neq 0$ and $\Delta = b^2 - 4ac \ge 0$, the usual quadratic formula is

$$x_{\pm} = \frac{-b \pm \sqrt{\Delta}}{2a}.$$

If $b < 0$ and $b^2 \gg |4ac|$, evaluating $(-b - \sqrt{\Delta})/(2a)$ may suffer cancellation. Rationalization gives

$$\frac{-b - \sqrt{\Delta}}{2a} = \frac{2c}{-b + \sqrt{\Delta}},$$

provided the denominator is nonzero. The analogous rationalization applies to the other root when $b > 0$. Moreover, forming b^2 can overflow even if the roots are finite. For $b \neq 0$, the identity

$$\sqrt{b^2 - 4ac} = |b|\sqrt{1 - \frac{4ac}{b^2}}$$

illustrates a rescaling that avoids directly squaring a large b; its implementation must also guard the intermediate quotient. When $|4ac/b^2| \ll 1$, the Taylor approximation $\sqrt{1-t} = 1 - \frac{1}{2}t + O(t^2)$ explains the sensitivity of the small root. In particular, for $b^2 \gg |4ac|$,

$$x_- \approx \begin{cases} -b/a + c/b, & b > 0, \\ -c/b, & b < 0. \end{cases}$$

Higher-order terms follow from $\sqrt{1-t} = 1 - \frac{1}{2}t - \frac{1}{8}t^2 + O(t^3)$.

Example 4.11. Two Ways to Compute a Mean
For $a, b \in \mathbb{F}$ compare

$$\widehat{m} = (a \oplus b) \oslash 2, \qquad \widehat{m}' = (a \oslash 2) \oplus (b \oslash 2).$$

If $|a + b| \le F_{\max}$, the first is correctly rounded:

$$\widehat{m} = \mathrm{rnd}\left(\frac{a+b}{2}\right), \qquad \min(a, b) \le \widehat{m} \le \max(a, b).$$

It may nevertheless overflow when $|a + b|$ exceeds the finite range, even though the mean is representable. The second avoids such intermediate overflow; it is correctly rounded when the inputs are sufficiently large, and otherwise has ULP error at most 1.

Theorem 4.12. Correctly Rounded Mean
Let $C \in [2^{e_{\min}+M+3}, 2^{e_{\max}}] \cap \mathbb{F}$. For $a, b \in \mathbb{F}$, define

$$m(a, b) = \begin{cases} (a \oplus b) \oslash 2, & |a| < C \text{ and } |b| < C, \\ (a \oslash 2) \oplus (b \oslash 2), & \text{otherwise.} \end{cases}$$

Then, assuming the stated floating-point operations are defined,

$$m(a, b) = \mathrm{rnd}\left(\frac{a+b}{2}\right) \qquad (a, b \in \mathbb{F}).$$

5 Error-Free Computations

Fix $\mathbb{F} = \mathbb{F}_{E,M}$ and $\mathrm{rnd} = \mathrm{rnd}_{E,M}$ as before. For $a \in \mathbb{F} \setminus \{0\}$ let $e_a \in [e_{\min}, e_{\max}] \cap \mathbb{Z}$ denote its floating-point exponent, taking $e_a = e_{\min}$ for subnormals. Set $e_0 = e_{\min}$ when an exponent is needed for zero. Write $\Delta_{\min} \coloneqq 2^{e_{\min}-M}$ for the least positive subnormal spacing. All exactness assertions below concern finite results with the indicated range conditions.

Lemma 5.1. Integer Significands
A real number a is in $\mathbb{F}$ if and only if there exist $m, e \in \mathbb{Z}$ with

$$a = m2^{e-M}, \qquad |m| < 2^{M+1}, \qquad e_{\min} \le e \le e_{\max}.$$

In particular, every $a \in \mathbb{F}$ is an integer multiple of $\Delta_{\min}$.

Proof. For normal values take $e = e_a$ and let m be the integer encoded by the $(M+1)$ significand bits; for subnormals take $e = e_{\min}$ and $|m| < 2^M$. Conversely, for any displayed pair (m, e), a significand with at most $M+1$ bits is either normal at a suitable exponent or subnormal after shifting to $e_{\min}$. If a shift would exceed $e_{\max}$, the bound $|m| < 2^{M+1}$ prevents it. □

Theorem 5.2. Hauser: Exact Addition
For $a, b \in \mathbb{F}$,

$$|a + b| \le 2^{e_{\min}+1} \implies a \oplus b = a + b.$$

Proof. Both a and b are integer multiples of $\Delta_{\min}$, hence so is $a + b$. Every such multiple in $[-2^{e_{\min}+1}, 2^{e_{\min}+1}]$ is representable. Therefore $a + b \in \mathbb{F}$ and rounding does nothing. □

Theorem 5.3. Sterbenz: Exact Subtraction
For $a, b \in \mathbb{F}$,

$$\frac{a}{2} \le b \le 2a \implies a \ominus b = a - b.$$

Equivalently, one may assume $b/2 \le a \le 2b$.

Proof. The hypothesis forces $a, b \ge 0$. The zero case is immediate. By exchanging a, b and changing the sign if necessary, assume $a \ge b$, so $b \le a \le 2b$. Write $a = m_a 2^{e_a - M}$ and $b = m_b 2^{e_b - M}$ by the lemma. As $a \ge b$, we have $e_a \ge e_b$ (with the convention for subnormals), and thus

$$a - b = (m_a 2^{e_a - e_b} - m_b) 2^{e_b - M} = m' 2^{e_b - M}, \qquad m' \in \mathbb{Z}.$$

Since $0 \le a - b \le b$, one has $0 \le m' \le m_b < 2^{M+1}$, so $a - b \in \mathbb{F}$. Thus $a \ominus b = a - b$. □

Proposition 5.4. Exact Scaling and Multiplication
Suppose $a, b \in \mathbb{F} \setminus \{0\}$ have $|a| = (\alpha_0.\alpha_1 \ldots \alpha_M)_2 \, 2^{e_a}$ and $|b| = 2^{e_b}$. If $e_a + e_b \in [e_{\min}, e_{\max}]$, then

$$a \otimes b = ab.$$

Similarly, if $e_a - e_b \in [e_{\min}, e_{\max}]$, then $a \oslash b = a/b$. More generally, a product is exact whenever its significand has at most $M+1$ significant binary digits and its exponent lies in the representable range.

Remark. Sterbenz's theorem does not preclude catastrophic cancellation: subtraction of nearby *represented* floats may be exact, while an error inherited from earlier rounded operations becomes large relative to the small difference.

Theorem 5.5. Dekker: Exact Addition Residual
For $a, b \in \mathbb{F}$ with $|a+b| \le F_{\max}$, put

$$s = a \oplus b, \qquad r = (a+b) - s.$$

Then $r \in \mathbb{F}$. If $e_a \ge e_b$, the algorithm

$$\begin{aligned} s &= a \oplus b, \\ b' &= s \ominus a, \qquad \textit{(Fast2Sum)} \\ z &= b \ominus b' \end{aligned}$$

returns $(s, z) = (a \oplus b, r)$; in particular, $a + b = s + z$ exactly.

Sketch. The integer-significand representation shows that the exact sum and its rounded value have a common binary grid at the smaller operand's exponent. Their difference has at most $M+1$ significant bits, hence $r \in \mathbb{F}$. For Fast2Sum, it suffices to verify

$$s - a \in \mathbb{F}, \qquad b - (s \ominus a) \in \mathbb{F}.$$

Under $e_a \ge e_b$ these exactness properties imply $b' = s - a$ and $z = b - b' = r$. The two auxiliary verifications are the steps isolated for Homework 2 in the lecture notes. □

Theorem 5.6. FMA Recovery of a Product Residual
For $a, b \in \mathbb{F}$ with $|ab| \le F_{\max}$, let $s = a \otimes b$ and $r = ab - s$. If

$$e_a + e_b \ge e_{\min} + M,$$

then $r \in \mathbb{F}$ and

$$r = \mathrm{fma}(a, b, -s).$$

Remark. The exponent condition controls whether the low-order bits of the exact product fit in the floating-point format. The corresponding failure at the weaker exponent bound $e_a + e_b = e_{\min} + M - 1$ is examined in Homework 2.

6 Automatic Error Analysis

Fix $N \in \mathbb{N}$ input variables and consider expressions generated by

$$e ::= c \mid x_i \mid -e \mid e + e \mid e \times e, \qquad c \in \mathbb{F}, \quad i \in \{1, \dots, N\}.$$

Only expressions whose floating-point evaluation does not overflow are considered. The expression e is syntax; its *semantics* is a function of the input valuation $\sigma = (\sigma_1, \dots, \sigma_N) \in \mathbb{F}^N$.

Definition 6.1. Floating-Point and Real-Valued Semantics
The *floating-point semantics* and *real-valued semantics* of an expression e are functions

$$[\![e]\!] \colon \mathbb{F}^N \to \mathbb{F}, \qquad [\![e]\!]_{\mathrm{r}} \colon \mathbb{F}^N \to \mathbb{R},$$

recursively defined by

$$\begin{aligned} [\![c]\!](\sigma) &= c, & [\![x_i]\!](\sigma) &= \sigma_i, \\ [\![-e]\!](\sigma) &= -[\![e]\!](\sigma), & [\![e_1 + e_2]\!](\sigma) &= [\![e_1]\!](\sigma) \oplus [\![e_2]\!](\sigma), \\ [\![e_1 \times e_2]\!](\sigma) &= [\![e_1]\!](\sigma) \otimes [\![e_2]\!](\sigma). \end{aligned}$$

The real-valued semantics uses the same rules with $+$ and $\times$ in place of $\oplus$ and $\otimes$, respectively.

Example 6.2. Different Semantics
For $e = (1 + x_1) + (-x_1)$, if $2^{M+1} \in \mathbb{F}$ and $\sigma_1 = 2^{M+1}$, then ties-to-even gives

$$[\![e]\!](\sigma) = 0, \qquad [\![e]\!]_{\mathrm{r}}(\sigma) = 1.$$

Thus the same expression has different floating-point and real-valued evaluations.

Definition 6.3. Automatic Range and Error Analysis
Let $l_i, r_i \in \mathbb{F}$ with $l_i \le r_i$, and fix the set of admissible input valuations

$$\Sigma := \prod_{i=1}^{N} ([l_i, r_i] \cap \mathbb{F}) \subseteq \mathbb{F}^N.$$

A *sound range analysis* produces intervals $\mathrm{range}(e) \subseteq \mathbb{R}$ such that

$$[\![e]\!](\sigma) \in \mathrm{range}(e) \qquad (\forall \sigma \in \Sigma).$$

A *sound absolute-error analysis* produces $\mathrm{err}(e) \in \mathbb{R}_{\ge 0}$ such that

$$|[\![e]\!](\sigma) - [\![e]\!]_{\mathrm{r}}(\sigma)| \le \mathrm{err}(e) \qquad (\forall \sigma \in \Sigma).$$

The dependence on the fixed Σ is suppressed in this notation.

Proposition 6.4. Recursive Range Analysis
For each subexpression e, set

$$\begin{aligned} \operatorname{range}(c) &= [c, c], \qquad \operatorname{range}(x_i) = [l_i, r_i], \\ \operatorname{range}(-e) &= [-b, -a] \qquad\qquad \textit{if } \operatorname{range}(e) = [a, b]. \end{aligned}$$

If $\operatorname{range}(e_i) = [a_i, b_i]$ *for* $i = 1, 2$, *define*

$$\operatorname{range}(e_1 + e_2) \coloneqq [\operatorname{rnd}(a_1 + a_2), \operatorname{rnd}(b_1 + b_2)].$$

These rules are sound, assuming no overflow in the analysis itself.

Proof. The constant, variable and negation cases follow directly from the semantics. By induction, $[\![e_i]\!](\sigma) \in [a_i, b_i]$, so

$$a_1 + a_2 \le [\![e_1]\!](\sigma) + [\![e_2]\!](\sigma) \le b_1 + b_2.$$

Apply monotonicity of rnd to obtain the asserted enclosure for $[\![e_1 + e_2]\!](\sigma)$. □

Proposition 6.5. Recursive Absolute-Error Analysis
Set

$$\operatorname{err}(c) = \operatorname{err}(x_i) = 0, \qquad \operatorname{err}(-e) = \operatorname{err}(e).$$

If $\operatorname{range}(e_i) = [a_i, b_i]$ *for* $i = 1, 2$, *define*

$$\begin{aligned} \operatorname{err}(e_1 + e_2) \coloneqq\ & \operatorname{err}(e_1) + \operatorname{err}(e_2) \\ & + \frac{1}{2} \operatorname{ulp}(\max\{|a_1 + a_2|, |b_1 + b_2|\}) \,. \end{aligned}$$

Under the same no-overflow assumptions, these rules are sound.

Proof. Put $u_i = [\![e_i]\!](\sigma)$ and $v_i = [\![e_i]\!]_{\mathrm{r}}(\sigma)$. Then

$$\begin{aligned} |\operatorname{rnd}(u_1 + u_2) - (v_1 + v_2)| &\le |\operatorname{rnd}(u_1 + u_2) - (u_1 + u_2)| + |u_1 - v_1| + |u_2 - v_2| \\ &\le \tfrac{1}{2} \operatorname{ulp}(u_1 + u_2) + \operatorname{err}(e_1) + \operatorname{err}(e_2). \end{aligned}$$

Since $u_i \in [a_i, b_i]$, $|u_1 + u_2| \le \max\{|a_1 + a_2|, |b_1 + b_2|\}$; ulp is nondecreasing in absolute value, proving the claim. The remaining cases are immediate. □

Remark. Limitations of Interval Analysis. The rules bound subexpressions independently and need not preserve dependencies between occurrences of the same variable. For $e = x_1 - x_1$ (shorthand for $x_1 + (-x_1)$) and $x_1 \in [-1, 1]$, ordinary interval arithmetic yields $[-2, 2]$, although both semantics evaluate to 0. Likewise, a positive error bound can be returned even when the actual rounding error vanishes. Refining the analysis to account for dependencies is a separate problem. The multiplication and square-root analysis rules discussed as exercises in the lecture are addressed in Homework 2, not assumed as proved results here.